

MATH 2270 — Linear Algebra

Practice Test 3 — Eigenvalues, Inner Products & Orthogonality

ANSWER KEY

Unit 3: §5.1–5.3, 5.5 (Eigenvalues, Diagonalization, Markov Chains) and §6.1–6.5, 7.1 (Inner Products, Gram–Schmidt, Least Squares, Orthogonal Matrices)

Name: _____

Date: _____

Practice test — twice as long as the real test (200 points).

The real Test 3 is 100 points in 50 minutes, so allow about 100 minutes for this whole practice test, or work through half of each section in 50 minutes. None of these questions appear on the real test. Same rules as the real test: closed book, closed notes, and a row-reduction calculator or software is allowed for Section 6 ONLY.

Section 1 — True / False

36 points

Circle **T** or **F** for each statement.

1. If A is a triangular matrix, then its eigenvalues are its diagonal entries. **TRUE**
 $\det(A - \lambda I)$ is the product of the diagonal entries of $A - \lambda I$.
2. The zero vector can be an eigenvector of a matrix. **FALSE**
 Eigenvectors are nonzero by definition (otherwise every λ would work).
3. Every orthogonal set of nonzero vectors in an inner product space is linearly independent. **TRUE**
 Take the inner product of a dependence relation with each vector.
4. If a square matrix is diagonalizable, then it is invertible. **FALSE**
 $\text{diag}(1, 0)$ is diagonalizable but singular.
5. If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable. **TRUE**
 Eigenvectors for distinct eigenvalues are independent, giving n independent eigenvectors.
6. The rotation of $\mathbb{R}^2$ by 90° has no real eigenvalues. **TRUE**
 Its characteristic polynomial is $\lambda^2 + 1$.
7. Eigenvectors belonging to different eigenvalues of a matrix are linearly independent. **TRUE**
 This is a standard theorem used in the diagonalization criterion.
8. If A and B are similar, then A and B have the same eigenvectors. **FALSE**
 They share eigenvalues, but eigenvectors change: $\mathbf{v}$ becomes $P^{-1}\mathbf{v}$.
9. For a regular Markov chain, the steady-state vector does not depend on the initial state vector $\mathbf{x}_0$. **TRUE**
 $P^k \mathbf{x}_0$ converges to the same $\mathbf{q}$ for every probability vector $\mathbf{x}_0$.

10. In every real inner product space, $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$. **TRUE**
 This is the symmetry axiom.
11. In every inner product space, $\|\mathbf{u} + \mathbf{v}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$. **FALSE**
 Only the triangle inequality $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ holds; equality needs parallel vectors in the same direction.
12. In every inner product space, $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$. **TRUE**
 The Cauchy–Schwarz inequality.
13. For a subspace W , the only vector in both W and $W^\perp$ is $\mathbf{0}$. **TRUE**
 A vector in both is orthogonal to itself, so $\langle \mathbf{v}, \mathbf{v} \rangle = 0$.
14. In Gram–Schmidt, the second vector $\mathbf{u}_2$ is orthogonal to $\mathbf{v}_1$. **TRUE**
 $\mathbf{u}_1 = \mathbf{v}_1$ and $\mathbf{u}_2 \perp \mathbf{u}_1$ by construction.
15. If A has linearly independent columns and $A = QR$ comes from Gram–Schmidt, then R is invertible. **TRUE**
 R is upper triangular with nonzero diagonal entries (the lengths of the $\mathbf{u}_i$).
16. The least-squares solution of $A\mathbf{x} = \mathbf{b}$ is always unique. **FALSE**
 It is unique only when the columns of A are independent.
17. Adding an x^2 column to a line fit (so fitting a parabola) can never increase the sum of squared errors. **TRUE**
 Every line is a parabola with $c_2 = 0$, so the minimum can only go down or stay the same.
18. If Q is a 5×3 matrix with orthonormal columns, then $QQ^T = I_5$. **FALSE**
 QQ^T has rank 3, so it cannot be I_5 ; it is the projection onto $\text{col}(Q)$. (But $Q^TQ = I_3$.)

Section 2 — Multiple Choice

48 points

Circle the letter of the best answer.

1. If λ is an eigenvalue of A , then the eigenvectors for λ , together with $\mathbf{0}$, form:
 (A) a single vector
 (B) the whole space $\mathbb{R}^n$
 (C) **a subspace called the eigenspace of λ**
 (D) a set that is not closed under addition
Answer: (C) — It is the null space of $A - \lambda I$, hence a subspace.
2. A real 2×2 matrix has characteristic polynomial $\lambda^2 + 4$. Then:
 (A) its eigenvalues are 2 and -2
 (B) it is singular
 (C) **it has no real eigenvectors**
 (D) it is the identity matrix
Answer: (C) — The roots are $\pm 2i$, which are not real, so there are no real eigenvectors.

3. Which condition guarantees that a 4×4 matrix is diagonalizable?

- (A) **it has four distinct eigenvalues**
- (B) it has no zero entries
- (C) its determinant is nonzero
- (D) all of its eigenvalues are real

Answer: (A) — Distinct eigenvalues give four independent eigenvectors. Nonzero determinant or real eigenvalues alone do not suffice.

4. Which of these matrices is *not* diagonalizable?

- (A) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$
- (B) $\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$
- (C) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- (D) **$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$**

Answer: (D) — $\lambda = 1$ has algebraic multiplicity 2 but only a 1-dimensional eigenspace. The others have enough eigenvectors.

5. Which of these stochastic matrices is regular?

- (A) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- (B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- (C) $\begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$
- (D) **$\begin{bmatrix} 0.3 & 0.6 \\ 0.7 & 0.4 \end{bmatrix}$**

Answer: (D) — All entries are positive. The others have zero entries in every power (identity, periodic swap, absorbing state).

6. On $\mathbb{R}^2$ let $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + 3u_2v_2$. Which vector is orthogonal to $(3, 1)$?

- (A) $(1, 3)$
- (B) **$(1, -1)$**
- (C) $(-1, 3)$
- (D) $(3, 1)$

Answer: (B) — $\langle (3, 1), (1, -1) \rangle = 3 - 3 = 0$. The others give 12, 6, and 12.

7. In an inner product space, $\|\mathbf{u}\| = 2$ and $\|\mathbf{v}\| = 3$. The largest possible value of $|\langle \mathbf{u}, \mathbf{v} \rangle|$ is:

- (A) 5
- (B) 1
- (C) 9
- (D) **6**

Answer: (D) — Cauchy–Schwarz: $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq 2 \cdot 3 = 6$, with equality when the vectors are parallel.

8. If W is a plane through the origin in $\mathbb{R}^3$ (with the dot product), then $W^\perp$ is:

- (A) a plane through the origin
- (B) **a line through the origin**
- (C) just the origin
- (D) all of $\mathbb{R}^3$

Answer: (B) — $\dim W + \dim W^\perp = 3$, so $\dim W^\perp = 1$.

9. In Gram–Schmidt with $\mathbf{u}_1 = \mathbf{v}_1$, the next orthogonal vector is $\mathbf{u}_2 =$

- (A) $\mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1$
- (B) $\mathbf{v}_2 - \frac{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle}{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle} \mathbf{u}_1$
- (C) $\mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{v}_2$
- (D) $\frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1$

Answer: (A) — Subtract the projection of $\mathbf{v}_2$ onto $\mathbf{u}_1$; the last option is the projection itself, not what is left.

10. If $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an *orthonormal* basis of an inner product space, the coordinate of $\mathbf{v}$ along $\mathbf{u}_2$ is:

- (A) $\langle \mathbf{v}, \mathbf{u}_2 \rangle$
- (B) $\langle \mathbf{u}_2, \mathbf{u}_2 \rangle$
- (C) $\|\mathbf{v}\|$
- (D) $\langle \mathbf{v}, \mathbf{v} \rangle$

Answer: (A) — Orthonormality gives $c_i = \langle \mathbf{v}, \mathbf{u}_i \rangle$.

11. The normal equations $A^T A \mathbf{x} = A^T \mathbf{b}$ always:

- (A) have no solution when $A \mathbf{x} = \mathbf{b}$ is inconsistent
- (B) **have at least one solution**
- (C) have exactly one solution
- (D) have only the solution $\mathbf{x} = \mathbf{0}$

Answer: (B) — They are consistent because the projection of $\mathbf{b}$ onto $\text{col}(A)$ always exists. The solution is unique only if the columns of A are independent.

12. If the columns of A are linearly dependent, then the least-squares solutions of $A \mathbf{x} = \mathbf{b}$:

- (A) do not exist
- (B) are unique
- (C) **are infinitely many, all with the same $A \hat{\mathbf{x}}$**
- (D) equal the exact solutions

Answer: (C) — $A^T A$ is singular, so $\hat{\mathbf{x}}$ is not unique, but the projection $A \hat{\mathbf{x}}$ is unique.

13. To fit $y = a + bx$ to the points $(1, 3), (2, 5), (4, 9)$, the matrix A and vector $\mathbf{b}$ are:

- (A) $A = \begin{bmatrix} 1 & 3 \\ 1 & 5 \\ 1 & 9 \end{bmatrix}$, $\mathbf{b} = (1, 2, 4)$
- (B) $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 4 & 1 \end{bmatrix}$, $\mathbf{b} = (3, 5, 9)$
- (C) $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix}$, $\mathbf{b} = (3, 5, 9)$
- (D) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}$, $\mathbf{b} = (3, 5, 9)$

Answer: (C) — Columns: ones and the x -values; right-hand side: the y -values.

14. The least-squares line is the line that minimizes:

- (A) **the sum of the squared vertical distances to the points**
- (B) the sum of the absolute vertical distances
- (C) the largest vertical distance
- (D) the sum of the squared perpendicular distances

Answer: (A) — Vertical residuals, squared and added: the SSE.

15. If Q is an orthogonal matrix, then Q^{-1} equals:

- (A) Q
- (B) $-Q$
- (C) Q^T
- (D) Q^2

Answer: (C) — $Q^T Q = I$ means the transpose is the inverse.

16. Let Q be 5×3 with orthonormal columns. Which statement is true?

- (A) $QQ^T = I_5$
- (B) $Q^T Q = I_3$
- (C) Q is an orthogonal matrix
- (D) $\det Q = 1$

Answer: (B) — $Q^T Q$ lists the dot products of the columns. Q is not square, so it has no determinant and is not called an orthogonal matrix.

Section 3 — Short Answer / Definitions

32 points

Fill in each blank with the best term.

- The polynomial $\det(A - \lambda I)$ is called the characteristic polynomial of A .
- The set of all eigenvectors for λ , together with $\mathbf{0}$, is called the eigenspace of λ .
- The number of times λ appears as a root of the characteristic polynomial is its algebraic multiplicity.
- The dimension of the eigenspace of λ is its geometric multiplicity.
- A matrix A is diagonalizable if $A = PDP^{-1}$ for some invertible P and diagonal D .

6. For a Markov chain with transition matrix P , a probability vector $\mathbf{q}$ with $P\mathbf{q} = \mathbf{q}$ is called the steady-state vector.
7. In an inner product space, the norm of $\mathbf{v}$ is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.
8. The distance between $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.
9. The inequality $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ is called the Cauchy–Schwarz inequality.
10. The algorithm that turns a linearly independent set into an orthogonal set spanning the same subspace is called the Gram–Schmidt process.
11. A basis whose vectors are pairwise orthogonal is called an orthogonal basis.
12. The equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ are called the normal equations.
13. The vector $\mathbf{b} - A \hat{\mathbf{x}}$ is called the residual vector.
14. In a line fit, the matrix whose columns are the constant column $\mathbf{1}$ and the x -values is called the design matrix.
15. A square matrix Q with $Q^T Q = I$ is called an orthogonal matrix.
16. Writing $A = QR$ with Q having orthonormal columns and R upper triangular is called the QR decomposition.

Section 4 — Quick Computations

24 points

No software needed for this section — these should be fast. Show minimal work.

1. Find all eigenvalues of $\begin{bmatrix} -1 & 0 & 0 \\ 4 & 6 & 0 \\ 2 & 3 & -5 \end{bmatrix}$.

Answer: $-1, 6, -5$

Lower triangular: the eigenvalues are the diagonal entries.

2. Is $\mathbf{v} = (1, 1)$ an eigenvector of $A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$? If so, find its eigenvalue.

Answer: Yes, $\lambda = 3$

$A\mathbf{v} = (3, 3) = 3\mathbf{v}$.

3. A 3×3 matrix has the eigenvalue 5 with algebraic multiplicity 3 and an eigenspace of dimension 2. Is the matrix diagonalizable? Why?

Answer: No

It has only 2 independent eigenvectors (geometric multiplicity $2 < 3$), so it is defective. *Grading: 1 pt answer, 2 pts reason.*

4. On $\mathbb{R}^2$ let $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 4u_2 v_2$. Find the cosine of the angle between $\mathbf{u} = (1, 0)$ and $\mathbf{v} = (1, 1)$.

Answer: $\frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$

$\langle \mathbf{u}, \mathbf{v} \rangle = 1$, $\|\mathbf{u}\| = 1$, $\|\mathbf{v}\| = \sqrt{1+4} = \sqrt{5}$.

5. In M_{22} with $\langle U, V \rangle = \text{tr}(U^T V)$ (the sum of products of corresponding entries), find $\langle A, B \rangle$ and $\|A\|$ for $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$.

Answer: $\langle A, B \rangle = 2$ and $\|A\| = \sqrt{6}$

$\langle A, B \rangle = 3 + 0 + 0 - 1 = 2$; $\|A\|^2 = 1 + 4 + 0 + 1 = 6$. *Grading: 2 pts + 1 pt.*

6. On $C[0, 1]$ with $\langle f, g \rangle = \int_0^1 f g \, dx$, are $f(x) = x$ and $g(x) = x - \frac{2}{3}$ orthogonal?

Answer: Yes

$\int_0^1 x(x - \frac{2}{3}) \, dx = \frac{1}{3} - \frac{1}{3} = 0$.

7. Find the least-squares solution $\hat{x}$ of $\begin{bmatrix} 1 \\ 2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and the length of the residual.

Answer: $\hat{x} = 2$, $\|\mathbf{r}\| = \sqrt{5}$

$A^T A = 5$, $A^T \mathbf{b} = 10$, so $\hat{x} = 2$. Residual $\mathbf{b} - A\hat{x} = (2, -1)$, length $\sqrt{5}$. *Grading: 2 pts + 1 pt.*

8. The first two columns of a 3×3 orthogonal matrix are $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$ and $(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3})$. Find a possible third column.

Answer: $\pm (-\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$

It must be a unit vector orthogonal to both (e.g. the cross product): $(-2, 2, 1)/3$ has length 1 and dot products 0 with both columns.

Section 5 — Inner Product or Not?

12 points

(a) (6 points)

On P_2 (polynomials of degree at most 2) define $\langle p, q \rangle = p(0)q(0) + p(1)q(1)$. Is this an inner product? Justify: if it fails, name the axiom and give a specific counterexample; if it holds, verify all the axioms.

Not an inner product. Symmetry and linearity hold, but positivity (definiteness) fails: $p(x) = x^2 - x = x(x - 1)$ is not the zero polynomial, yet $p(0) = p(1) = 0$ gives $\langle p, p \rangle = 0$. *Grading: 2 pts symmetry/linearity, 4 pts counterexample.*

(b) (6 points)

On $\mathbb{R}^2$ define $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_1 v_2 + u_2 v_1 + 2u_2 v_2$. Is this an inner product? Justify: if it fails, name the axiom and give a counterexample; if it holds, verify all the axioms.

It is an inner product. *Symmetry:* swapping $\mathbf{u} \leftrightarrow \mathbf{v}$ gives the same expression. *Linearity:* each term is linear in $\mathbf{u}$ (and in $\mathbf{v}$). *Positivity:* $\langle \mathbf{u}, \mathbf{u} \rangle = u_1^2 + 2u_1 u_2 + 2u_2^2 = (u_1 + u_2)^2 + u_2^2 \geq 0$, and it equals 0 only if $u_2 = 0$ and $u_1 = -u_2 = 0$, i.e. $\mathbf{u} = \mathbf{0}$. *Grading: 1 pt symmetry, 2 pts linearity, 3 pts positivity.*

Section 6 — Computation with Software**48 points**

You may use a row-reduction calculator or software for this section. Show the matrix or system you set up and row reduced, and clearly state your final answer. Box or circle your final answers.

Problem 1 — Diagonalization**(6 points)**

Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ -3 & 1 & -1 \end{bmatrix}$. **(a)** Find the characteristic polynomial and eigenvalues (2 pts). **(b)** Find an eigenvector for each eigenvalue (2 pts). **(c)** Give P and D with $A = PDP^{-1}$ (2 pts).

(a) Expanding along column 3: $\det(A - \lambda I) = (-1 - \lambda)[(2 - \lambda)(1 - \lambda)]$, so $\lambda = 2, 1, -1$ (all distinct). **(b)** Row reducing $A - \lambda I$: $\lambda = 1$: $(-1, 1, 2)$; $\lambda = 2$: $(-1, 0, 1)$; $\lambda = -1$: $(0, 0, 1)$ (any nonzero multiples). **(c)** With the order $1, 2, -1$: $P = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 1 \end{bmatrix}$, $D = \text{diag}(1, 2, -1)$.

Problem 2 — A Defective Matrix**(6 points)**

Let $B = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 5 \end{bmatrix}$. **(a)** Find the eigenvalues and their algebraic multiplicities (2 pts). **(b)** Find a basis for the eigenspace of $\lambda = 3$ (2 pts). **(c)** Is B diagonalizable? Explain (2 pts).

(a) Triangular: $\lambda = 3$ (algebraic multiplicity 2) and $\lambda = 5$ (1). **(b)** $B - 3I = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 2 \end{bmatrix}$ row reduces to a matrix of rank 2, so the null space is spanned by $(1, 0, 0)$. **(c)** No: the geometric multiplicity of $\lambda = 3$ is $1 < 2$, so B has only 2 independent eigenvectors in total.

Problem 3 — Weighted Gram–Schmidt**(6 points)**

On $\mathbb{R}^3$ let $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + 3u_2v_2 + 2u_3v_3$, $\mathbf{v}_1 = (1, 0, 1)$, $\mathbf{v}_2 = (2, 1, 0)$. **(a)** Use Gram–Schmidt to find an orthogonal pair $\mathbf{u}_1, \mathbf{u}_2$ (4 pts). **(b)** Find the orthonormal pair $\mathbf{q}_1, \mathbf{q}_2$ (2 pts).

(a) $\mathbf{u}_1 = (1, 0, 1)$, $\langle \mathbf{u}_1, \mathbf{u}_1 \rangle = 1 + 0 + 2 = 3$, $\langle \mathbf{v}_2, \mathbf{u}_1 \rangle = 2$. $\mathbf{u}_2 = \mathbf{v}_2 - \frac{2}{3}\mathbf{u}_1 = (\frac{4}{3}, 1, -\frac{2}{3})$ (or $(4, 3, -2)$). Check: $\langle \mathbf{u}_1, (4, 3, -2) \rangle = 4 + 0 - 4 = 0$. **(b)** $\|\mathbf{u}_1\| = \sqrt{3}$; for $(4, 3, -2)$: $16 + 27 + 8 = 51$. $\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1, 0, 1)$, $\mathbf{q}_2 = \frac{1}{\sqrt{51}}(4, 3, -2)$.

Problem 4 — Gram–Schmidt and QR**(6 points)**

Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 2 & 0 \end{bmatrix}$. (a) Use Gram–Schmidt with the dot product on the columns of A to find orthonormal $\mathbf{q}_1, \mathbf{q}_2$ (3 pts). (b) Find $R = Q^T A$ so that $A = QR$ (2 pts). (c) Check that $Q^T Q = I$ (1 pt).

(a) $\|\mathbf{a}_1\| = 3$, $\mathbf{q}_1 = \frac{1}{3}(1, 2, 2)$. $\mathbf{a}_2 \cdot \mathbf{q}_1 = \frac{1}{3}(3 + 6) = 3$; $\mathbf{u}_2 = \mathbf{a}_2 - 3\mathbf{q}_1 = (3, 3, 0) - (1, 2, 2) = (2, 1, -2)$, $\|\mathbf{u}_2\| = 3$, $\mathbf{q}_2 = \frac{1}{3}(2, 1, -2)$. (b) $R = \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$. (c) $\mathbf{q}_1 \cdot \mathbf{q}_1 = \mathbf{q}_2 \cdot \mathbf{q}_2 = 1$ and $\mathbf{q}_1 \cdot \mathbf{q}_2 = \frac{1}{9}(2 + 2 - 4) = 0$.

Problem 5 — Best-Fit Line**(6 points)**

Find the least-squares line $y = a + bx$ through $(1, 6), (2, 5), (3, 3), (4, 2), (5, 1)$. (a) Compute $A^T A$ and $A^T \mathbf{b}$ (2 pts). (b) Solve for a, b (2 pts). (c) Predict y at $x = 6$ (1 pt), and say what the slope tells you (1 pt).

(a) $A^T A = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}$, $A^T \mathbf{b} = (17, 38)$. (b) $(a, b) = (7.3, -1.3)$, so $y = 7.3 - 1.3x$. (c) At $x = 6$: $7.3 - 7.8 = -0.5$. The slope -1.3 : y decreases by about 1.3 for each unit increase in x . (SSE = 0.3, not asked.)

Problem 6 — Best-Fit Parabola**(6 points)**

Fit $y = c_0 + c_1x + c_2x^2$ by least squares to $(-2, 0), (-1, 1), (0, -1), (1, -1), (2, 6)$. (a) Write A (1 pt). (b) Compute $A^T A$ and $A^T \mathbf{b}$ (2 pts). (c) Solve for c_0, c_1, c_2 (2 pts). (d) Predict y at $x = 3$ (1 pt).

(a) Rows $(1, x, x^2)$ for $x = -2, -1, 0, 1, 2$. (b) $A^T A = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix}$, $A^T \mathbf{b} = (5, 10, 24)$. (c) $(c_0, c_1, c_2) = (-1, 1, 1)$, so $y = -1 + x + x^2$. (d) At $x = 3$: $-1 + 3 + 9 = 11$. (SSE = 10, not asked.)

Problem 7 — Markov Chain**(6 points)**

A Markov chain has transition matrix $P = \begin{bmatrix} 0.6 & 0.1 & 0.2 \\ 0.3 & 0.7 & 0.2 \\ 0.1 & 0.2 & 0.6 \end{bmatrix}$ and initial state $\mathbf{x}_0 = (0.5, 0.5, 0)$. (a) Is P stochastic? Why? (1 pt) (b) Find $\mathbf{x}_1$ and $\mathbf{x}_2$ (2 pts). (c) Find the steady-state vector $\mathbf{q}$ (3 pts).

(a) Yes: entries are nonnegative and each column sums to 1. (b) $\mathbf{x}_1 = P\mathbf{x}_0 = (0.35, 0.5, 0.15)$; $\mathbf{x}_2 = P\mathbf{x}_1 = (0.29, 0.485, 0.225)$. (c) Row reduce $(P - I)\mathbf{q} = \mathbf{0}$: the solution is a multiple of $(8, 14, 9)$. Dividing by the sum 31: $\mathbf{q} = \left(\frac{8}{31}, \frac{14}{31}, \frac{9}{31}\right) \approx (0.258, 0.452, 0.290)$.

Problem 8 — An Orthogonal Complement**(6 points)**

Let $W = \text{span}\{(1, 1, 0, 1), (0, 1, 1, -1)\} \subseteq \mathbb{R}^4$ with the dot product. **(a)** Set up the system whose solutions are the vectors in $W^\perp$ (2 pts). **(b)** Row reduce it and find a basis for $W^\perp$ (3 pts). **(c)** What is $\dim W^\perp$? (1 pt)

(a) $\mathbf{x} \in W^\perp$ iff $x_1 + x_2 + x_4 = 0$ and $x_2 + x_3 - x_4 = 0$, i.e. $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix} \mathbf{x} = \mathbf{0}$. **(b)** RREF: $\begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}$, so $x_1 = x_3 - 2x_4$, $x_2 = -x_3 + x_4$. Basis: $(1, -1, 1, 0)$ and $(-2, 1, 0, 1)$. **(c)** $\dim W^\perp = 2$ (indeed $2 + 2 = 4$).